8

FURTHER APPLICATIONS OF INTEGRATION

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8.2

Area of a Surface of Revolution

In this section, we will learn about:

The area of a surface curved out by a revolving arc.

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- **A surface of revolution is formed when a curve is rotated about a line.**

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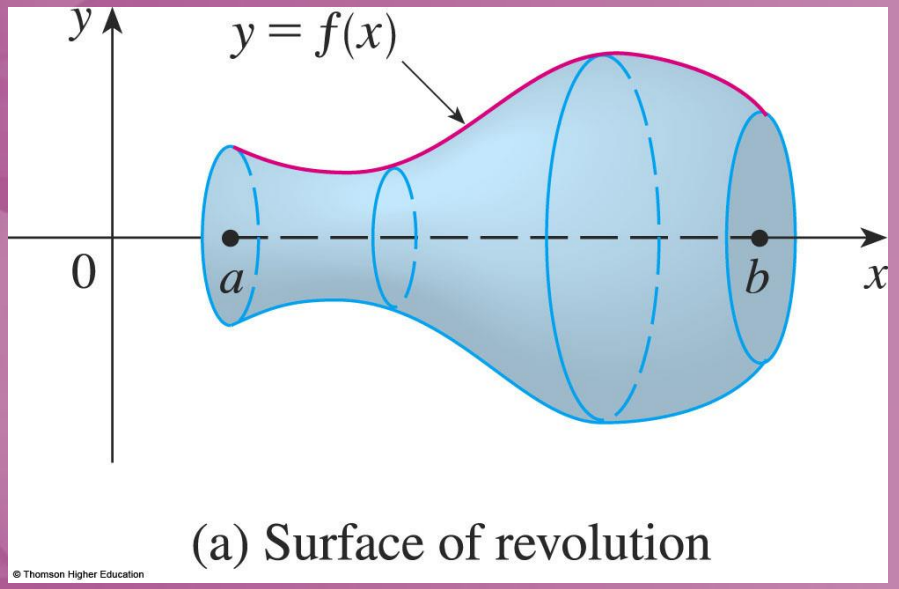
- We want to define the area of a surface of revolution in such a way that it corresponds to our intuition.

  - If the surface area is *A*, we can imagine that painting the surface would require the same amount of paint as does a flat region with area *A* .

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SURFACE AREA

- Consider the surface shown here.
  - It is obtained by rotating the curve $y = f(x)$, $a \le x \le b$, about the x-axis, where f is positive and has a continuous derivative.



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Formula 4

- Thus, in the case where f is positive and has a continuous derivative, we define the surface area of the surface obtained by rotating the curve $y = f(x)$, $a \leq x \leq b$, about the x-axis as:

$$S = \int_a^b 2\pi R(x)\sqrt{1+\left[f'(x)\right]^2}\,dx$$

- ***Where R is the distance between the curve and axis of rotation***

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SURFACE AREA **Formula 6**

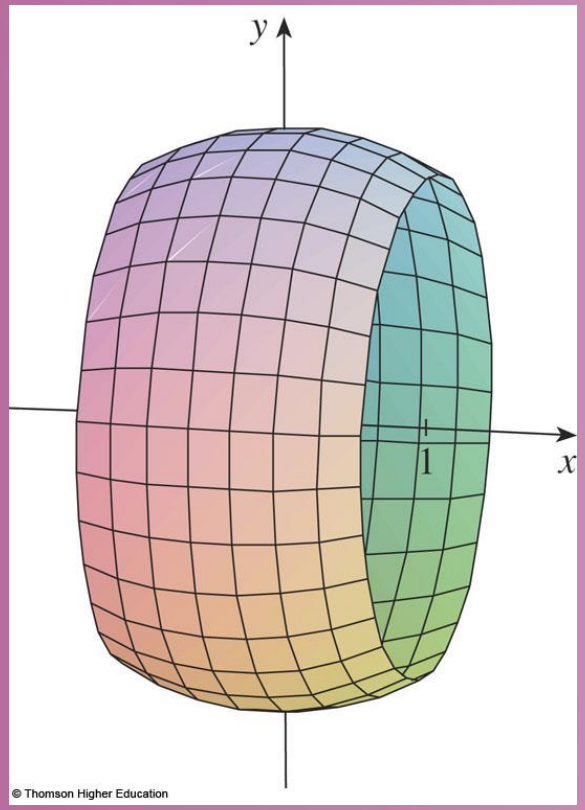
- If the curve is described as $x = g(y)$, $c \le y \le d$, then the formula for surface area becomes:

$$S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dx$$

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Example 1

- The curve $y = \sqrt{4 - x^2}$, $-1 \le x \le 1$, is an arc of the circle $x^2 + y^2 = 4$.
- Find the area of the surface obtained by rotating this arc about the x-axis.
  - The surface is a portion of a sphere of radius 2.

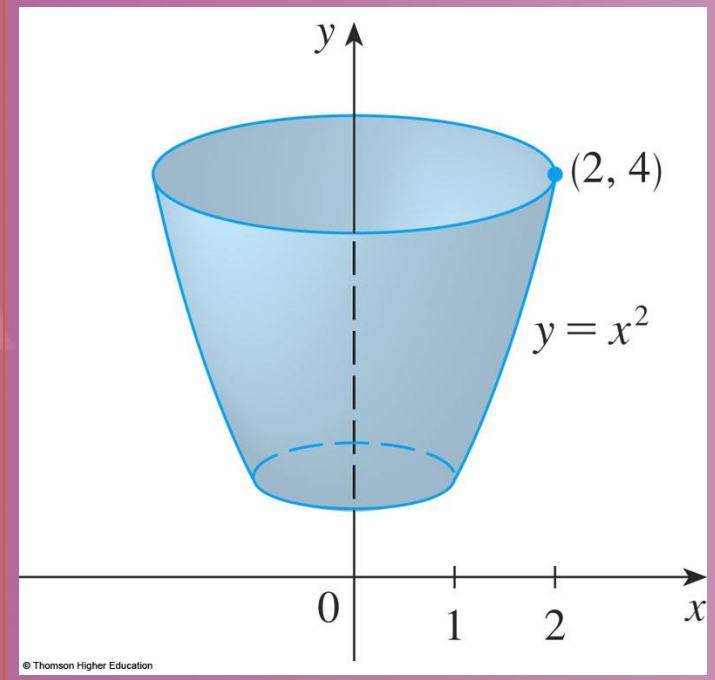


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Example 2

- The arc of the parabola $y = x^2$ from (1, 1) to (2, 4) is rotated about the y-axis.

- Find the area of the resulting surface.



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SURFACE AREA

E. g. 2—Solution 1

- Using $y = x^2$ and $dy/dx = 2x$, from Formula 8, we have:

$$S = \int 2\pi x \, ds$$

$$= \int_1^2 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

$$= 2\pi \int_1^2 x\sqrt{1 + 4x^2} \, dx$$

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SURFACE AREA

E. g. 2—Solution 1

- Substituting $u = 1 + 4x^2$, we have $du = 8x\,dx$.

- Remembering to change the limits of integration, we have:

$$S = \frac{\pi}{4}\int_5^{17} \sqrt{u}\,du = \frac{\pi}{4}\left[\tfrac{2}{3}u^{3/2}\right]_5^{17}$$

$$= \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$$

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SURFACE AREA

E. g. 2—Solution 2

- Using

$$x = \sqrt{y} \quad \text{and} \quad dx/dy = \frac{1}{2\sqrt{y}},$$

- we have the following solution.

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SURFACE AREA

E. g. 2—Solution 2

$$S = \int 2\pi x\,ds = \int_1^4 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy$$

$$= 2\pi \int_1^4 \sqrt{y}\sqrt{1 + \frac{1}{4y}}\,dy$$

$$= \pi \int_1^4 \sqrt{4y+1}\,dy$$

$$= \frac{\pi}{4}\int_5^{17} \sqrt{u}\,du \qquad (\text{where } u = 1 + 4y)$$

$$= \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$$

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SURFACE AREA

Example 3

- Find the area of the surface generated by rotating the curve $y = e^x$, $0 \le x \le 1$, about the x-axis.

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SURFACE AREA

Example 3

- Using Formula 5 with $y = e^x$ and $dy/dx = e^x$, we have:

$$S = \int_0^1 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= 2\pi \int_0^1 e^x \sqrt{1 + e^{2x}}\, dx$$

$$= 2\pi \int_1^e \sqrt{1 + u^2}\, du \quad (\text{where } u = e^x)$$

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SURFACE AREA

Example 3

$$= 2\pi \int_{\pi/4}^{\alpha} \sec^3\theta \, d\theta \quad (\text{where } u = \tan\theta \text{ and } \alpha = \tan^{-1} e)$$

$$= 2\pi \tfrac{1}{2}\left[\sec\theta\tan\theta + \ln\left|\sec\theta + \tan\theta\right|\right]_{\pi/4}^{\alpha} \quad (\text{E.g.8, Sec.7.2})$$

$$= \pi\left[\sec\alpha\tan\alpha + \ln\left(\sec\alpha + \tan\alpha\right) - \sqrt{2} - \ln\left(\sqrt{2}+1\right)\right]$$

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Example 3

- Since tan $\alpha = e$, we have:
  - $sec^2\alpha = 1 + \tan\alpha = 1 + e^2$

- Thus,

$$S = \pi\left[e\sqrt{1+e^2} + \ln\left(e+\sqrt{1+e^2}\right) - \sqrt{2} - \ln\left(\sqrt{2}+1\right)\right]$$

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